

RESEARCH INTERESTS

- Computer Systems and Architecture
- Hardware-Software Co-Design for Database Systems on Processing-in-Memory

EDUCATION

Yonsei University, Seoul, Republic of Korea

2022.03 - Present

MSc/PhD Student in Computer Science

- Advisor: Prof. Youngsok Kim
- Research area: System Software, Database System, Processing-in-Memory

Yonsei University, Seoul, Republic of Korea

2017.03 - 2022.02

BSc in Computer Science

PUBLICATIONS

1. **Suhyun Lee**, Hanna Cha, Chaemin Lim, Hanjun Kim, and Youngsok Kim, “MEPID-Join: Scaling Processing-in-DIMM Joins Using Adaptive Key-Level Replication.” *In Proc. 53rd International Conference on Very Large Data Bases (VLDB)*, 2027 (to appear).
2. Sungchul Lee, **Suhyun Lee**, Hanna Cha, Chaemin Lim, and Youngsok Kim. “GJP: Fast and Efficient Join-Project Query Processing on GPUs Using Density-Aware Matrix Multiplication.” *In Proc. 2027 ACM International Conference on Management of Data (SIGMOD)*, 2027 (to appear).
3. Jinwoo Choi, **Suhyun Lee**, and Youngsok Kim. “RecoNN: A Resource Contention-Aware Runtime for Latency-Sensitive On-Device Inference.” *In Proc. 44th IEEE International Conference on Computer Design (ICCD)*, 2026 (to appear).
4. Chaemin Lim, **Suhyun Lee**, Jinwoo Choi, Kwanghyun Park, Jinho Lee, Joonsung Kim, and Youngsok Kim. “FaScalSQL: A Fast and Scalable GPU-Accelerated SQL Query Engine for Out-of-Memory Tables.” *In Proc. 42nd IEEE International Conference on Data Engineering (ICDE)*, 2026.
5. Chaemin Lim, **Suhyun Lee**, Jinwoo Choi, Joonsung Kim, Jinho Lee, Youngsok Kim. “DMO-DB: Mitigating the Data Movement Bottlenecks of GPU-Accelerated Relational OLAP.” *In Proc. 34th International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2025.
6. **Suhyun Lee**, Chaemin Lim, Jinwoo Choi, Heelim Choi, Chan Lee, Yongjun Park, Kwanghyun Park, Hanjun Kim, and Youngsok Kim. “SPID-Join: A Skew-resistant Processing-in-DIMM Join Algorithm Exploiting the Bank- and Rank-level Parallelisms of DIMMs.” *In Proc. 2025 ACM International Conference on Management of Data (SIGMOD)*, 2025.
7. Chaemin Lim, **Suhyun Lee**, Jinwoo Choi, Jounghoo Lee, Seongyeon Park, Hanjun Kim, Jinho Lee, and Youngsok Kim. “Design and Analysis of a Processing-in-DIMM Join Algorithm: A Case Study with UPMEM DIMMs.” *In Proc. 2023 ACM International Conference on Management of Data (SIGMOD)*, 2023.
8. Jinwoo Choi, Jaeyeon Kim, Chaemin Lim, **Suhyun Lee**, Jinho Lee, Dokyung Song, and Youngsok Kim. “GuardiaNN: Fast and Secure On-Device Inference in TrustZone Using Embedded SRAM and Cryptographic Hardware.” *In Proc. 23rd ACM/IFIP International Middleware Conference (Middleware)*, 2022.
9. Jounghoo Lee, Yeonan Ha, **Suhyun Lee**, Jinyoung Woo, Jinho Lee, Hanhwi Jang, and Youngsok Kim. “GCoM: a detailed GPU core model for accurate analytical modeling of modern GPUs.” *In Proc. 49th Annual International Symposium on Computer Architecture (ISCA)*, 2022.

PATENTS	Skew Resistance Processing in DIMM Device and Processing Method	
	<i>Youngsok Kim, Suhyun Lee, Chaemin Lim, Jinwoo Choi, and Hanjun Kim.</i>	
	JP Patent 7,803,601	
	Artificial Intelligence Device Based on Trust Environment	
	<i>Youngsok Kim, Jinwoo Choi, Chaemin Lim, Suhyun Lee, Dokyung Song, and Jinho Lee</i>	
PROJECTS	KR Patent 10-2,946,913	
	Artificial Intelligence Device Based on Trust Environment	
	<i>Youngsok Kim, Jinwoo Choi, Chaemin Lim, Suhyun Lee, Dokyung Song, and Jinho Lee</i>	
	JP Patent 7,621,675	
	Apparatus and Method for Analyzing Graphics Processor Performance based on Graphics Processor Analytical Model	
AWARDS AND HONORS	<i>Youngsok Kim, Jounghoo Lee, Yeonan Ha, Suhyun Lee</i>	
	US Patent 12,223,561	
	Optimizing Database Systems Using Processing-in-Memory Hardware	
	<i>National Research Foundation of Korea (NRF)</i>	2024.09 - 2026.08
	Exploring Quality-Aware NN Scheduler	
EXPERIENCE	<i>LG Electronics</i>	2024.03 - 2025.12
	• Scholarship NRF Fellowship Program for Doctoral Students.	
	2024.06	
	• Best Poster Award Computer System Society Academic Conference	
	2023.02	
EXPERIENCE	Teaching Assistant: <i>Computer Architecture (CSI3102),</i>	
	<i>Logic Circuit Design (CSI2111),</i>	2023.03 - 2023.07
	<i>Information Security (CSI4109)</i>	2022.09 - 2022.12
		2022.03 - 2022.07